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Editor-in-Chief
Surface Topography: Metrology and Properties

Dear Editor,

I am pleased to submit the manuscript "Optical Profilometry Versus the Stylus: A Spectral Theory for Closing the Accuracy Gap" for consideration as a research paper in Surface Topography: Metrology and Properties.

Why STMP? STMP is the natural home for this work because the manuscript addresses the journal's core remit—surface topography measurement and characterisation—at a fundamental level: it pairs tactile and optical measurement physics in one consistent k-space operator formalism and quantifies *how* and *why* the modalities produce different topographies from the same surface. This is not an incremental comparison study; it is a computational framework that connects instrument physics to measured surface properties, validated across 180 parameter combinations, six surface classes, an out-of-sample scaling-law check, and a public measured WLI transfer-function dataset (Hirose et al., Applied Optics 64:5461, 2025; CC-BY-4.0). The work speaks directly to STMP's readership of metrologists, instrument developers, and surface engineers who need to know when stylus and optical measurements can be trusted, when they disagree, and how to close the gap.

Novelty and contributions.

1. **SNR-dependent resolution–bandwidth trade-off.** A metrologically anchored, noise-generalised form of the Fourier uncertainty bound: $\Delta x_{\text{eff}} \cdot \Delta f_{\text{eff}} \geq (1/4\pi)(1 + 1/\overline{\text{SNR}})$, with operational definitions of the effective widths and an explicit falsification protocol, quantifying how finite signal-to-noise ratio inflates the classical Fourier uncertainty bound. It is instrument-independent and experimentally testable.
2. **Instrument transfer functions** $H(f_x, f_y)$ for stylus profilometry, white-light interferometry, and angle-resolved scatterometry, computed in one consistent simulation framework—tip dilation (with the empirical small-slope curvature-bias transfer replacing the classical Gaussian-tip idealisation), numerical aperture, wavelength, and noise. The optical OTF component is the standard incoherent OTF; its combination with the stylus channel in a single framework, the empirical dilation transfer, and the cross-modality k-space comparison are the novel elements. Validated through a 180-point parameter sweep, ensemble statistics ($n = 120$ across six surface classes), and a public measured WLI-ITF dataset (fitted cutoff within 15% of nominal).
3. **Reconstructability index** ρ—a single-number figure of merit (0–1) for

instrument capability, shown to be instrument-dominated (CV = 0.22 across four manufactured classes). Answers the practical question: "Can my instrument measure this surface?"

4. **Prescriptive gap-closure roadmap** identifying four specific optical parameter improvements (noise floor $\times 100$, NA 0.7–0.9, λ 400–450 nm, 2D phase unwrapping) that expand the stylus–optical equivalence regime from 10% (ensemble baseline) to 87% of k-space, with quantified technology readiness levels for each.
5. **Information-theoretic formulation of surface metrology** including mutual information $I(h; y)$, channel capacity C, and the fundamental MMSE bound on surface reconstruction—to the best of my knowledge the first application of Shannon theory to the comparison of different profilometry modalities.
6. **GUM-compliant uncertainty propagation** for the transfer function estimate, with dominant sources identified (tip radius: $\sim$60% of variance in H_{stylus}; numerical aperture: $\sim$55% in H_{optical}).

Context and positioning. The work builds on the PSD-based surface characterisation methodology established by Jacobs et al. (2017), the ITF framework of Foreman et al. (2013), Leach (2010), de Groot (2021), Dai et al. (2020), and Giusca et al. (2011), and the spectral stylus analyses of Wu (1999, 2000) and McCool (1984). It extends these approaches from single-instrument characterisation to the systematic cross-modality comparison that the surface metrology community has needed since Poon and Bhushan's early comparative work (1995). The information-theoretic formulation provides fundamental limits on surface reconstruction that complement existing ISO 25178 areal texture standards by quantifying *how much* information each modality extracts, not just *how accurately*.

Nature of the study. All results are computational; I do not have access to the multi-instrument laboratory facilities required for experimental validation. A set of falsifiable predictions and open challenges for experimental groups is therefore provided (Section 4.6), all feasible with standard laboratory equipment and commercially available calibration artefacts. As an independent check of the optical channel model, the manuscript benchmarks it against the public CC-BY-4.0 white-light-interferometry dataset of Hirose et al. (Applied Optics 64:5461, 2025; Zenodo 10.5281/zenodo.15750880), reproducing the measured instrument transfer functions within 15% of the nominal cutoff across three binary-pseudo-random-array samples. The complete physical framework (Hertzian contact mechanics, vector diffraction), scatterometry extension, six-class stress-test, GUM uncertainty analysis, and non-Gaussian ensemble validation are provided as Supplementary Material (19 pages). The manuscript is longer than a typical STMP research paper because it develops three interconnected frameworks—operator formalism, information theory, scatterometry extension—whose integration is essential to the central argument that k-space fusion weighted by local informa-

tion uncertainty is the optimal strategy for broadband surface metrology. The Supplementary Material separates extended derivations and stress-test details from the main narrative.

Additional information. This is a single-author study; the submission follows the journal's author-choice peer-review model in single-anonymous form, with author names and affiliations included. All simulation code and data will be deposited in a public repository (Zenodo) and cited by DOI upon acceptance, in accordance with IOP's research data policy. The manuscript has not been published previously and is not under consideration elsewhere. The author declares no competing interests.

I would welcome the opportunity to have this work reviewed by experts in surface topography measurement, optical instrumentation, and information theory.

Sincerely,

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